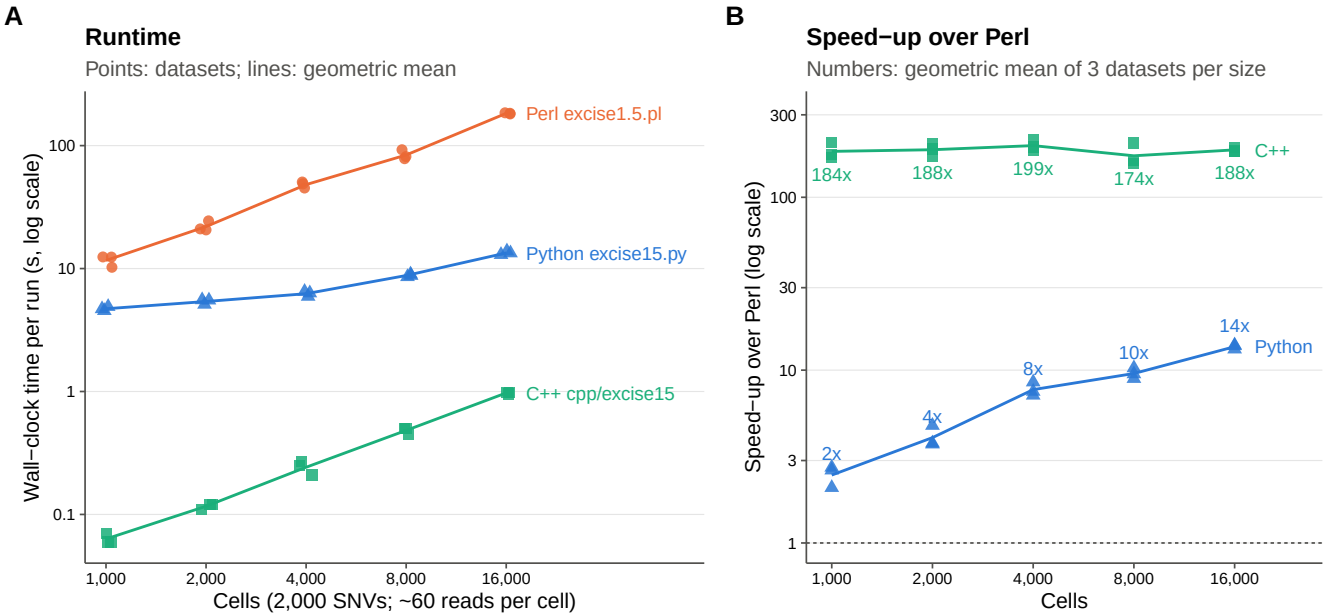

All tools: −t 4 −j 4 −pretrain on the same simulated BAMs (15 datasets), run one at a time. Times are full runs including BAM reading;
Python includes one−off numba compilation. Optimiser alone: Python 0.04−0.06 s, C++ at most 0.01 s per run. bc2xci identical to Perl: Python 10/15, C++ 12/15 ru
the rest reached different local optima of the stochastic search (1,000−2,000 cells; at most 1 cell labelled differently).